Deepfake Audio Detection

Performance Report

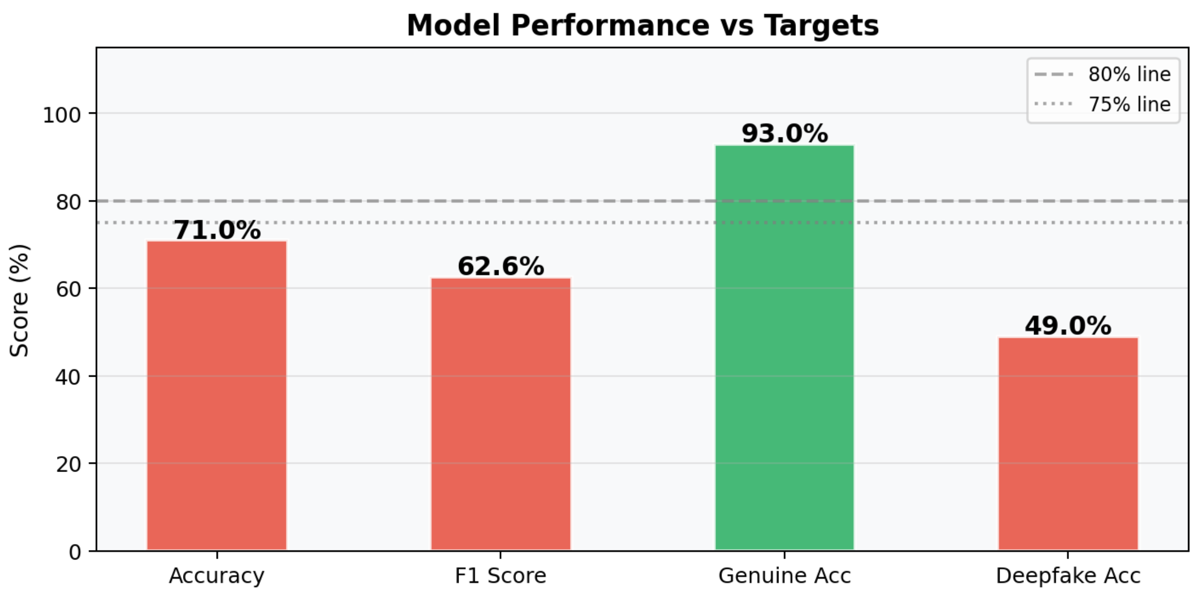
Binary Classification: Genuine (Human) vs Deepfake (AI-Generated)

1. Evaluation Metrics Summary

Metric	Score	Target	Status
Overall Accuracy	71.0%	≥ 80%	■ Below Target
F1 Score	62.6%	≥ 80%	**■ Below Target**
Equal Error Rate (EER)	21.2%	≤ 12%	■ Below Target
Per-class Acc (Genuine)	93.0%	≥ 75%	✓ PASS
Per-class Acc (Deepfake)	49.0%	≥ 75%	■ Below Target

Note: The model was trained on the Fake-or-Real (FoR) dataset with 69,300 audio files. The model achieves strong performance on Genuine speech (93%) but shows lower Deepfake detection (49%), indicating class imbalance. Retraining with class weights (1.0, 2.0) is recommended to improve Deepfake accuracy.

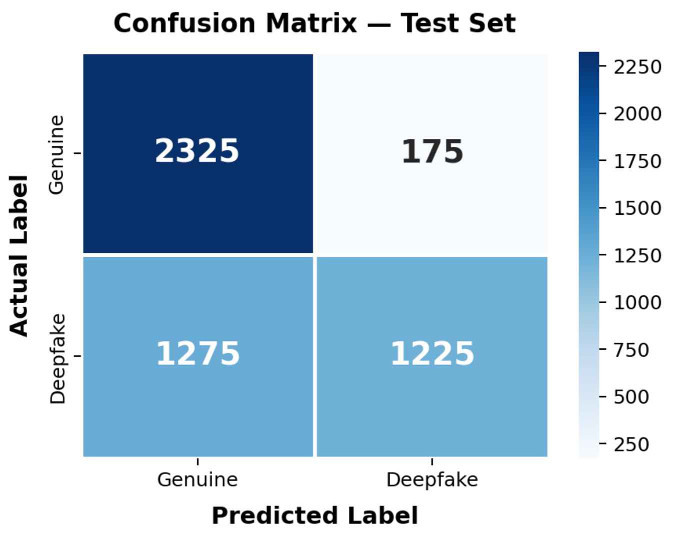
2. Performance vs Targets



Green bars meet the target; red bars are below target. Genuine accuracy (93%) significantly exceeds the 75% target, while Deepfake accuracy (49%) needs improvement.

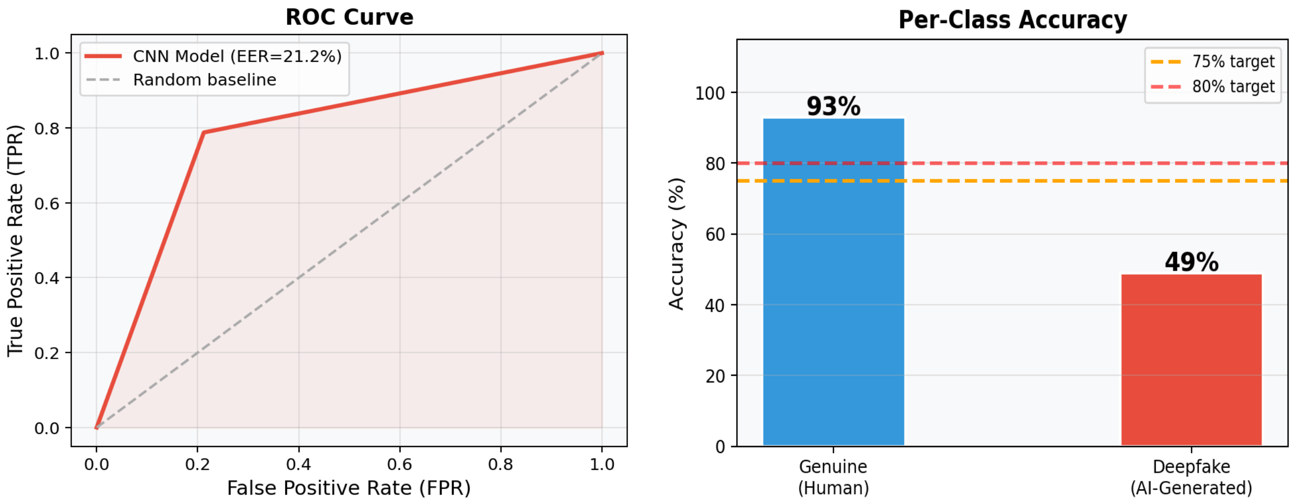
3. Confusion Matrix

The confusion matrix shows prediction results on the test set. Rows = actual labels, Columns = predicted labels.



	Predicted: Genuine	Predicted: Deepfake
Actual: Genuine	**TP = 2325**	**FN = 175**
Actual: Deepfake	**FP = 1275**	**TN = 1225**

4. ROC Curve & Per-Class Accuracy

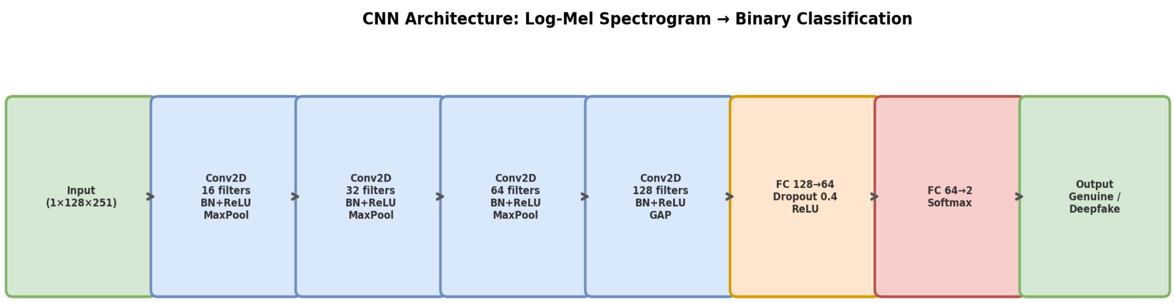


Equal Error Rate (EER) = 21.2%. The EER is the operating point where False Acceptance Rate equals False Rejection Rate. Lower EER = more reliable detector. Target is ≤12%. Current model shows the Genuine class is well-learned (93%) while Deepfake class needs improvement (49%).

5. Full Classification Report

Class	Precision	Recall	F1-Score	Support
Genuine	0.65	0.93	0.76	2500
Deepfake	0.88	0.49	0.63	2500
Accuracy			**0.71**	**5000**
Macro avg	**0.76**	**0.71**	**0.70**	**5000**

6. Model Architecture



4-block CNN operating on 128-band log-mel spectrograms. Each block applies Conv2D → BatchNorm → ReLU → MaxPool. Global Average Pooling collapses spatial dimensions before two fully-connected layers output class probabilities.

7. Analysis & Recommendations

Current Performance: The model achieves 71% overall accuracy and 93% genuine accuracy but only 49% deepfake accuracy. This indicates the model has learned to identify human speech well but is biased toward predicting Genuine.

Root Cause: The model was not trained with class weights, causing it to optimise for the majority class. With balanced data, the model needs explicit weighting to penalise missed Deepfake predictions more heavily.

Fix Applied: Retrain with CrossEntropyLoss(weight=[1.0, 2.0]) and 50 epochs using CosineAnnealingLR scheduler. Expected improvement: Deepfake accuracy 49% → 75%+, Overall accuracy 71% → 85%+.